

Weekly Report — Manual-Agent Benchmark: v2.1 to v2.2

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Scope: This week closed the OBD golden expert-review cycle and converted the manual lane's remaining stable failures into root-cause fixes — DTC-procedure navigation, corrupted section structure, and vehicle-identity delivery — each verified with a measured before/after on the live server, merged, and deployed to production.

1. Headline scores

Milestone	Mean overall (30 locked goldens)	Notes
v2.1 (2026-07-13)	0.777	Median 0.811, σ 0.195.
v2.2 (2026-07-21)	0.831	Median 0.887, σ 0.136, 30/30 above the pass floor — first all-green full run. Mean wall 72 s.

Same-rubric weekly delta: 0.777 $\rightarrow$ 0.831 (+0.054). Spread tightened by a third (σ 0.195 $\rightarrow$ 0.136) — the gain came from fixing the worst entries, not from inflating the top.

OBD lane milestone (same week): the expert-review cycle closed — 2 goldens revised per Deng Xiao's comments, the post-revision re-run held 15/15 pass, mean 0.938, and all 15 entries were promoted into the locked tier. Both eval lanes now run entirely on expert-approved locked goldens.

Per question type (v2.1 $\rightarrow$ v2.2)

Type	v2.1	v2.2
lookup	0.839	0.841
procedural	0.741	0.945
cross-section	0.752	0.792
image-required	0.779	0.815
adversarial	0.774	0.759

Procedural — the weakest category two weeks ago (0.567 at v2) — is now the strongest, on the strength of the DTC-navigation fixes below.

2. Problems found $\rightarrow$ fixes shipped $\rightarrow$ measured effect

Problem (short)	Issue / PR	Measured effect
OBD goldens carried 2 unresolved expert-review comments	#205 / PR #206, #207, #208	Both entries revised per review; re-run 15/15 pass holds (mean 0.938); all 15 promoted to the locked tier — OBD review cycle CLOSED
Judge could not see figures the agent surfaced (held last week)	#193 / PR #197	Merged after confound resolution; image-required 0.686 $\rightarrow$ 0.785 in the same-rubric re-run
DTC Quick Index mapped codes to nothing (occurrence counts only)	#210 / PR #211	Index now carries a per-code section slug column; dtc-001 0.400 $\rightarrow$ 0.988
P-code diagnostic sections invisible & sliced by junk NOTE-banner headings	#210 / PR #211	Banner filter extended; P0107 section recovered 112 $\rightarrow$ 15,264 chars; procedural-002 0.363 $\rightarrow$ 0.985
Vehicle identity reached the manual sub-agent only via LLM free text	#213 / PR #214 (closes #204)	Deterministic «VEHICLE» block injected from the session row (APP-60 fields + VIN); cross-004 0.347 $\rightarrow$ 0.755, zero wrong-manual reads across all verification runs
Prod deployment	both PRs live	Single-container redeploys $\times$ 2, warm LLM preserved (no cold-load window)

Supporting artifacts: every PR carries a branch-deploy verification report; raw run data committed under docs/eval-reports/ (harness24_manual_rerun_20260721.json, harness28_manual_baseline_20260721.json, harness29_vehicle_targeted.json, harness29_v2_2_baseline.json, harness27_obd_rerun.json).

3. Honest caveats on v2.2

- The 30/30 all-green is against the 0.4 regression floor; the working quality bar of 0.7 stands at 24/30. Both numbers are week-over-week highs (v2.1: 27/30 and 21/30 respectively).
- Run variance is reduced but not gone (σ 0.136): cross-003 swung 0.802 $\rightarrow$ 0.425 $\rightarrow$ 0.745 across three same-day runs with no related change. Milestone means are trustworthy; single entries still are not.
- Four stable low performers are now clearly isolated (they repeat their scores across runs, unlike the noise cases): lookup-005 (0.578, identical across 4 runs), image-006 (0.612 $\times$ 3), cross-006 (0.483–0.612), adversarial-006 (0.646–0.660). These are next week's root-cause targets.
- cross-004's residual gap (0.755 vs golden 0.95+) is the spec-lookup half of the question — manual selection is fixed, the cross-section spec retrieval is not.

4. Next steps

1. Root-cause the two zero-variance stable lows: lookup-005 and image-006 — their run-to-run score stability suggests structural causes (like the DTC-navigation class fixed this week), not model noise.
2. Characterise cross-006 and adversarial-006 (borderline stable-low vs variance).
3. cross-004 spec-lookup completion (second read into the engine-specs table).
4. Umbrella #116 continues: next milestone eval after the above lands becomes v2.3.